

10/7/26

Class Activity - 1

1. Student Academic Performance Dashboard.
 1. Qualitative and Quantitative variables.
Qualitative : Student ID, Department
Quantitative : Attendance (%), Internal marks.
2. Aesthetic mappings.
Position (X-axis) : Attendance
Position (Y-axis) : Internal marks
Color : Department.
3. Scale Types.
Attendance - continuous
Internal marks - continuous
Department - Discrete.
4. Why these mappings?
 - Position clearly shows the relationship between attendance and marks.
 - Color makes it easy to compare departments.
5. If shape was used instead of color.
Departments could still be identified but colours are easier to distinguish.
6. Best visualization.
Scatter plot, because it clearly shows the relationship between attendance and marks.

Scenario - 2:

1. Qualitative : vehicle model, manufacturer
Quantitative : Battery Capacity, price, Driving Range.
2. Aesthetic mappings:
X-axis : Battery capacity
Y-axis : price
Size : Driving Range
Color : Manufacturer

3. Scale types:

Battery capacity - continuous

Price - continuous

Driving Range - continuous

Manufacturer - Discrete.

4. Why these mappings?

Bubble size shows driving range

colour separates manufacturers.

5. If shape replaced color:

Manufacturers could still be identified, but colors are easier to compare.

6. Best visualization.

Bubble chart, because it represents four variables together.

Scenario-3:

1. Qualitative: Department

Quantitative: No. of patients, Average Treatment Cost

2. Aesthetic mappings:

X-axis: Department

Y-axis: No. of patients

Fill color: Department

Labels: Average Treatment cost.

3. Scale Types:

Department - Discrete

No. of patients - continuous

Treatment Cost - continuous.

4. Why these mappings.

Bars compare patient counts

Fill colours distinguish departments.

5. If shape replaced fill

Shapes are not suitable for bar charts.

6. Best visualization.

Bar chart, because it compares categories effectively.

Scenario - 4:

1. Qualitative: city

Quantitative: Data, Temperature

2. Aesthetic mappings:

X-axis: Date

Y-axis: Temperature

Color: city

Line Type: city

3. Scale Types:

Data - Continuous

Temperature - Continuous

city - Discrete

4. Why these mappings?

Different colors and line styles make cities easy to compare.

5. If shape replaced order.

Shapes are unsuitable for continuous line graphs.

6. Best visualisation:

Line chart, because it shows trends over time.

Scenario - 5:

1. Qualitative - month, Product category, Branch

Quantitative - Sales amount

2. Aesthetic Mappings:

X-axis: Product category

Y-axis: Sales amount

Fill color: Branch

3. Scale Types:

month - Ordinal

Product category - Discrete

Branch - Discrete

Sales Amount - Continuous

4. Why these mappings?

Fill color helps compare branches.

5. If shape replaced fill

Shapes cannot represent grouped bars effectively.

6. Best visualization.

Grouped Bar chart, because it compares categories across branches.

Scenario - 6 :

1. Qualitative - Employee ID, Department

Quantitative - Salary, Experience.

2. Aesthetic Mappings

X-axis : Salary

Y-axis : Frequency

3. Scale Types

Salary - continuous

Frequency - Continuous

4. Why these mappings

Histogram shows salary distribution

5. If color replaced bins

Color could not clearly show salary distribution.

6. Best visualisation.

Histogram because it displays frequency distribution.

Scenario - 7 :

1. Qualitative : Brand

Quantitative : RAM, Processor Speed, Price, Storage.

2. Aesthetic Mappings

X-axis : RAM

Y-axis : Price

Shape : Brand

Size : Storage

Color : Processor Speed.

3. Scale Types

RAM - continuous

Price - Continuous

Storage - continuous

Processor Speed - Continuous

Brand - Discrete

4. Why these mappings?
Multiple aesthetics represent multiple variables together.
5. If color replaced shape.
Brand identification may become confusing if processor speed also uses color.
6. Best visualization.
Scatter plot, because it compares several variables at once.

Scenario - 8:

1. Qualitative - Genre
2. Quantitative - Average Rating, No. of reviews, Release year.
2. Aesthetic Mappings
Position: Genre
Color Gradient: Average rating
3. Scale Types.
Genre - Discrete
Average Rating - Continuous
4. Why these mappings?
Color gradient highlights rating differences.
5. If shape replaced color
Ratings would be harder to interpret.
6. Best visualisation.
Heatmap, because it quickly identifies high-rated genres.

Scenario - 9:

1. Qualitative: State
Quantitative: Active cases, vaccination percentage, Population.
2. Aesthetic Mappings.
Position: State
Color Gradient: Active cases
3. Scale Types
State - Discrete
Active Cases - Continuous.

4. Why these mappings?

Sequential color clearly shows case intensity.

5. If shape replaced color

It would be difficult to compare the severity across states.

6. Best visualisation.

Choropleth Map, because it clearly shows geographical distribution.

Scenario - 10 :

1. Qualitative : Species

Quantitative : Sepal length, Sepal width, Petal length, Petal width.

2. Aesthetic Mappings :

X-axis : Sepal length

Y-axis : Petal length

color : Species

Shape : Species

Size : Sepal width.

3. Scale Type :

Sepal length - continuous

Petal length - continuous

Sepal width - continuous

Species - Discrete.

4. Why these mappings?

Color and shape distinguish species, while size represents sepal width.

5. If only shape was used.

Species could still be identified, but color improves readability.

6. Best visualization

Scatter plot, because it clearly shows clustering of different iris species.